

T. Chinoera

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 2 Theory

9189/2

JUNE 2015 SESSION

1 hour 15 minutes

Candidates answer on the question paper.

Additional materials:

Data Booklet

Mathematical Tables and/or Calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
TOTAL	

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Answer *all* questions

For
Examiner's
Use

- 1 (a) (i) State the ideal gas equation.

- (ii) Explain why real gases deviate from ideal behaviour.

- (iii) State **two** conditions necessary for a gas to approach ideal behaviour.

Explain your answer.

condition

explanation

condition

explanation

[7]

- (b) A 50.00 g sample of a volatile liquid was placed in a globe of volume 1 000 cm³. The globe was then heated to 91 °C so that all the liquid vaporised. The vapour exerted a pressure of 0.25 atm.

Calculate the relative molecular mass, M_r , of the liquid.

[1 atm = 101.3 kPa]

[2]

[Total:9]

- (a) (i) State Hess' Law.

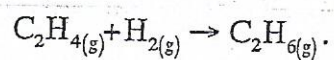
- (ii) Define the term *standard enthalpy change of combustion*.

- (iii) Table 2.1 shows enthalpy changes of two hydrocarbons.

Table 2.1

hydrocarbon	$\Delta H_c^\theta / \text{kJmol}^{-1}$
C_2H_6	1 559.8
C_2H_4	1 410.8

Calculate the enthalpy change of the reaction,



[7]

(b) State any **two** factors that affect the magnitude of lattice energy.

1. _____

2. _____

[2]

[Total:9]

3 (a) (i) Write the electronic configuration of magnesium.

(ii) Describe the reaction of magnesium with cold water.

(iii) State any **one** use of magnesium oxide.

[3]

(b) (i) Write an expression for the solubility product, K_{sp} , of calcium sulphate.

(ii) The solubility product of calcium sulphate is $2 \times 10^{-5} \text{ mol}^2 \text{ dm}^{-6}$ at 25°C .

Predict whether precipitation would occur if equal volumes of $1 \times 10^{-2} \text{ mol dm}^{-3} \text{ CaCl}_2$ and $1 \times 10^{-2} \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4$ are mixed.

[3]

(c) Explain why

(i) calcium sulphate is less soluble than magnesium sulphate,

(ii) magnesium carbonate decomposes at 400 °C while barium carbonate decomposes at 1 360 °C.

[6]

[Total:12]

4 (a) Fig. 4.1 shows the structure of isopentyl acetate, a honey bee pheromone.

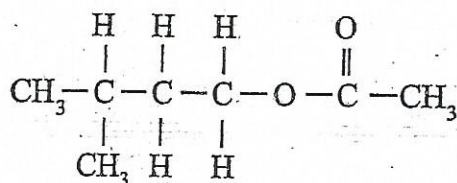


Fig.4.1

(i) Suggest why isopentyl acetate is used as a perfume.

- (ii) Draw and name the structures of the products of acid hydrolysis of isopentyl acetate.

[5]

- (b) (i) State the observations made when aqueous bromine is added to phenol.

- (ii) Draw the structure of the organic product in b(i).

[3]

[Total:8]

- 5 (a) Fig. 5.1 shows the structure of vitamin A.

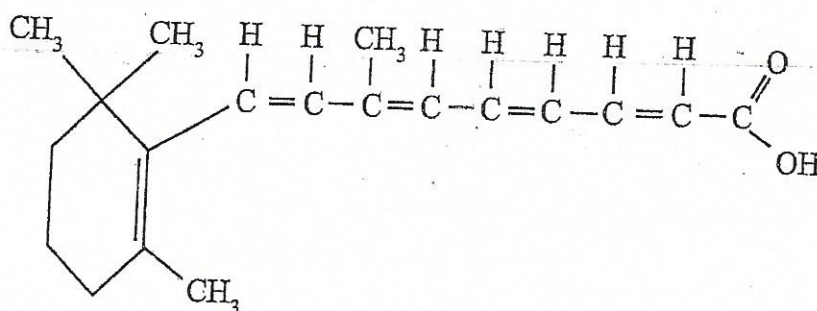


Fig. 5.1

- (i) Name any two functional groups in vitamin A.

1. _____

2. _____

(ii) Draw structures of the organic products formed when vitamin A is reacted with

1. HBr ,

2. HCN/NaOH ,

3. acidified $\text{K}_2\text{Cr}_2\text{O}_7$,

4. dinitrophenylhydrazine.

(b) The monomer for natural rubber is isoprene, $\text{CH}_2 = \text{C}(\text{CH}_3)\text{CH} = \text{CH}_2$.

(i) State the type of polymer that natural rubber shows.

(ii) Draw the structure of the repeat unit of natural rubber.

[2]

(c) Draw the structure of the organic product formed when isoprene reacts with

(i) cold dilute KMnO_4 ,

(ii) hot concentrated KMnO_4 .

[2]

[Total:10]